

SIMATS ENGINEERING
ASSESSMENT TOOL 2: Technical Presentation

COURSE NAME:	Cloud Computing and Big Data Analytics	COURSE CODE:	CSA1522
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Register Number	192372087

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)*

Assessment Tool Weightage: 10%

Dave's Taxonomy: Imitation (Level 1) Question Count: 5	Total Marks: 50
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Code of Conduct

I Atmakuri Sai Mrudula (192372087) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: A. Sai Mrudula

Assessment Tasks

Technical Presentation Topics

Prepare a 6-8 minute presentation on any one of the following topics. Your slides must include definitions, architecture, industry relevance, and one practical example.

1. Cloud computing characteristics and benefits for startup ecosystems.
2. How virtualization enabled the growth of cloud service providers.
3. A comparative presentation on IaaS, PaaS, and SaaS for an educational institution.
4. Web services in cloud platforms: SOAP vs REST for enterprise integration.
5. Why XaaS is becoming a dominant consumption model in modern IT.

Minimum slide flow:

- Title and objective
- Core technical explanation
- Architecture or process diagram
- Real-world use case
- Conclusion and key takeaway

Technical Presentation Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Technical Content	30%	Content is highly accurate, relevant, and insightful	Content is correct with minor gaps	Basic content with limited depth	Content is inaccurate or superficial
Structure and Flow	20%	Excellent logical flow and slide organization	Good structure with minor issues	Some organization but weak flow	Disorganized presentation
Use of Visuals	15%	Visuals strongly support technical communication	Relevant visuals used well	Limited or unclear visuals	No meaningful visuals
Communication Skills	20%	Confident, clear, and engaging delivery	Clear delivery with minor hesitation	Moderate clarity but uneven delivery	Weak delivery and low clarity
Professionalism	15%	Well-prepared and audience-focused presentation	Mostly professional	Basic preparation visible	Poor preparation and professionalism



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SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CO1-AT2-TECHNICAL PRESENTATION

CSA1522 : Cloud Computing for Bigdata Analytics

GUIDED BY:

Dr. P. Rajaram

PRESENTED BY :

Atmakuri Sai Mrudula (192372087)



A comparative presentation on IaaS, PaaS, and SaaS for an educational institution

OBJECTIVE

Cloud computing has transformed the way organizations use technology by providing computing resources over the internet. Instead of purchasing and maintaining expensive hardware and software, users can access services on demand. The three major cloud service models are Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS). Educational institutions use these models to improve teaching, learning, research, and administrative activities.

- Understand the three cloud service models.
- Compare IaaS, PaaS, and SaaS.
- Learn their architecture.
- Identify their importance in educational institutions.
- Explore real-world examples.



INTRODUCTION

Cloud service models define the level of control and responsibility shared between the cloud provider and the customer.

- **Infrastructure as a Service (IaaS)** provides virtualized computing resources such as servers, storage, networking, and operating systems. Users have maximum control over the infrastructure.
- **Platform as a Service (PaaS)** provides a complete development environment where developers can build, test, and deploy applications without managing servers.
- **Software as a Service (SaaS)** provides ready-to-use software applications through the internet. Users simply log in and use the application without worrying about installation or maintenance.



IaaS (Infrastructure as a Service)

Provides virtual machines, storage, networking, and servers.



PaaS (Platform as a Service)

Provides a platform for developing and deploying applications.



SaaS (Software as a Service)

Provides ready-to-use software through the internet.



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CORE TECHNICAL EXPLANATION



IaaS

- Virtual servers
- Storage
- Networking
- Scalability
- Pay-as-you-use pricing

PaaS

- Development tools
- Runtime environment
- Database services
- Application deployment

SaaS

- Web-based applications
- No installation required
- Automatic updates
- Accessible from anywhere

➤ Educational Example For IaaS

University hosts its student management system on cloud virtual machines.

➤ Educational Example PaaS

Students build web applications without configuring servers.

➤ Educational Example SaaS

Students use online learning platforms.



IaaS (Infrastructure as a Service)

- Virtual servers
- Storage
- Networking
- Scalability
- Pay-as-you-use pricing

Educational Example

University hosts its student management system on cloud virtual machines.



PaaS (Platform as a Service)

- Development tools
- Runtime environment
- Database services
- Application deployment

Educational Example

Students build web applications without configuring servers.



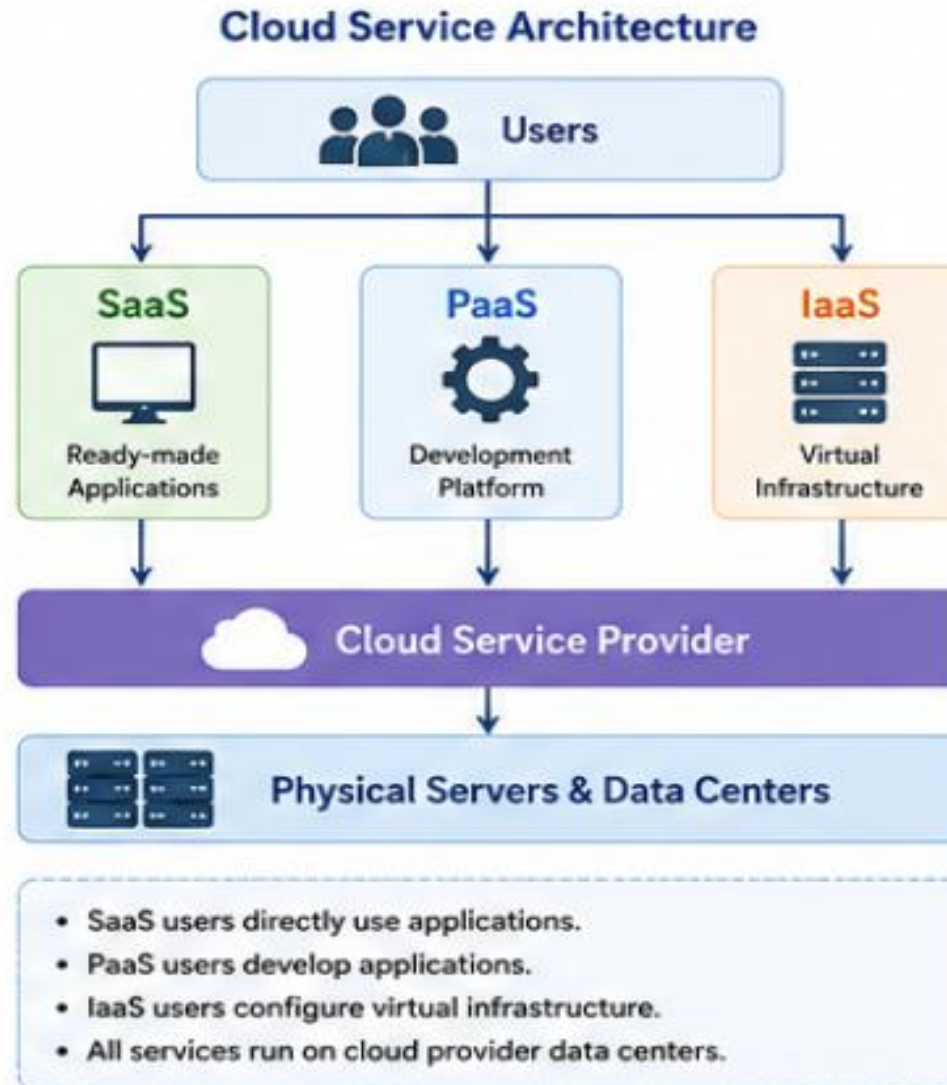
SaaS (Software as a Service)

- Web-based applications
- No installation required
- Automatic updates
- Accessible from anywhere

Educational Example

Students use online learning platforms.

ARCHITECTURE DIAGRAM



COMPARATIVE ANALYSIS

The three cloud service models differ mainly in the level of control provided to users.

Feature	IaaS	PaaS	SaaS
User Control	High	Medium	Low
Ease of Use	Moderate	Easy	Very Easy
Maintenance	User	Shared	Provider
Customization	High	Medium	Low
Cost	Moderate	Moderate	Low



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INDUSTRY RELEVANCE



Educational institutions increasingly rely on cloud computing because it reduces infrastructure costs and improves accessibility.

IaaS Applications

- Virtual computer laboratories
- Cloud storage for research data
- Disaster recovery systems
- Digital libraries

SaaS Applications

- Online classes
- Email services
- Video conferencing
- Assignment submission
- Document collaboration

PaaS Applications

- Student portal development
- Examination management systems
- Learning Management System (LMS) development
- Academic mobile applications





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Real-World Use Case



A modern university can use all three cloud service models simultaneously.

IaaS

The university hosts databases, virtual servers, and storage on cloud infrastructure.

PaaS

Developers create attendance systems, online examination portals, and student mobile applications using cloud development platforms.

SaaS

Students and faculty access cloud-based applications for online learning, email, virtual classrooms, document sharing, and assignment submission.

Benefits

- Better security
- Easy scalability
- Remote learning support
- Faster deployment of educational services



Conclusion

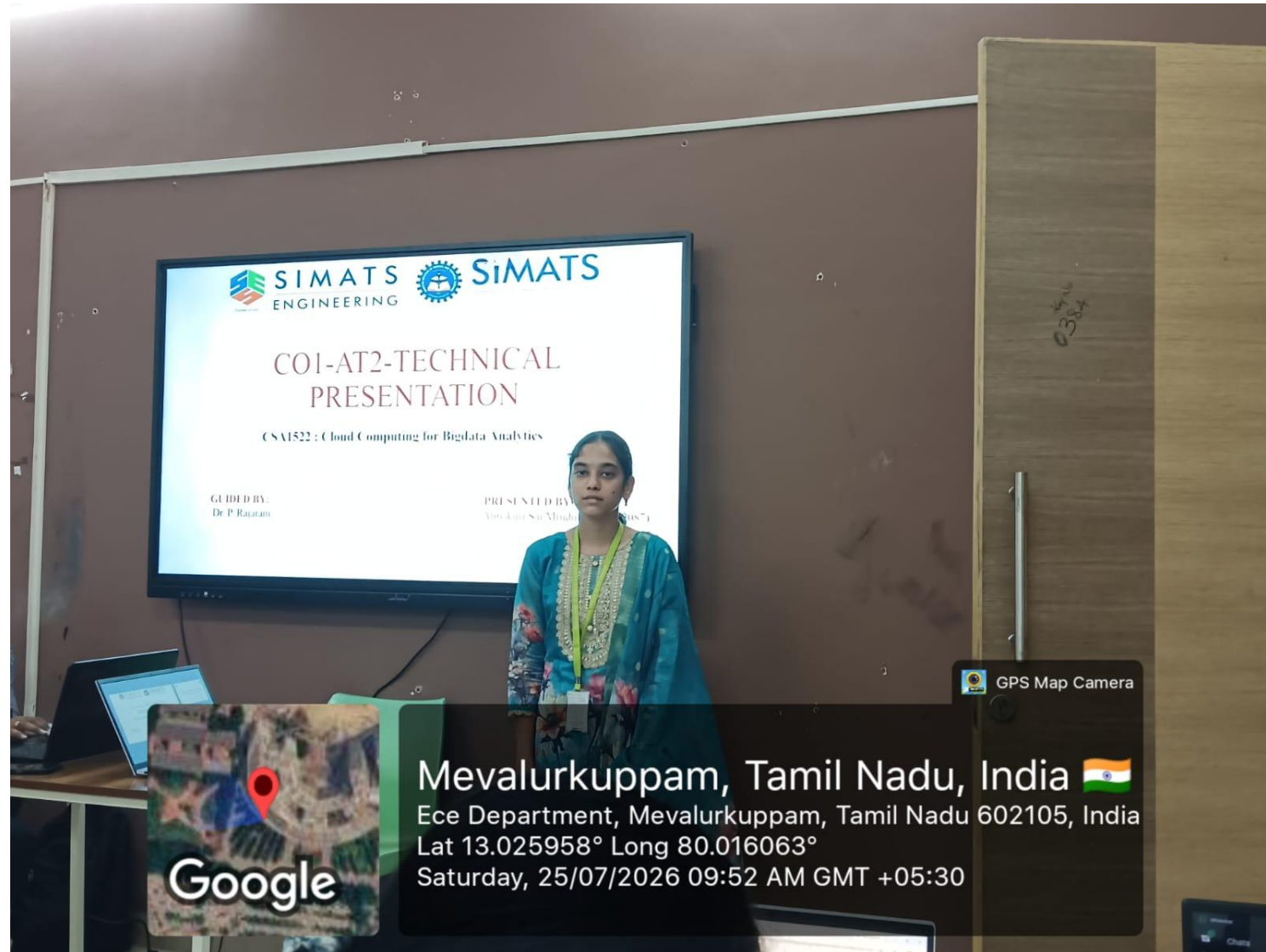


Cloud computing has become an essential technology in modern education. Each cloud service model serves a different purpose:

- **IaaS** provides flexible infrastructure for hosting educational systems.
- **PaaS** enables developers to build innovative educational applications.
- **SaaS** offers ready-to-use software for students, teachers, and administrators.

By combining all three models, educational institutions can achieve cost savings, improved efficiency, enhanced collaboration, and better learning experiences.

Geotag



THANK YOU